The Width of the Conformal Fan: Dependence and the Variance of Realized Coverage

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Draft — companion note

Abstract

Split conformal prediction's realized, calibration-conditional coverage fluctuates around the nominal $1-\alpha$. For exchangeable-but-independent calibration scores it follows the $\mathrm{Beta}(k, n-k+1)$ law, variance about $\alpha(1-\alpha)/n$, the classical fan. We show the width of that fan is governed by the sign of the cross-sample dependence. The mean stays pinned at the marginal level whatever the dependence; the variance, to leading order, is the average pairwise covariance of the exceedance indicators at the operating quantile times the independent fan. Positive (extendable) dependence adds a between-dataset term and widens the fan; negative dependence narrows it, to exactly zero at the maximally negatively associated contest floor $\rho = -1/(n-1)$, where the realized coverage equals the nominal level identically. We prove the positive side exactly through de Finetti and the law of total variance, the negative side exactly at the floor and to leading order in general, and, summed over all target levels, exactly: a convex-order contraction of the exceedance count against a concave functional shows negative association never widens the total fan. For the canonical equicorrelated Gaussian family the single level is exact and monotone, and dispersively so, via a strong-log-concavity argument and Tweedie's formula. The general single-level statement is left as a conjecture, supported numerically across dependence structures. The governing sign is the one the companion note attaches to the finite de Finetti measure: a genuine prior widens the fan, the signed corner collapses it.

1 Setup

Split conformal returns a set with marginal coverage $1-\alpha$. Fix the calibration set; the coverage you then realize on fresh test points is a random variable, and it is not $1-\alpha$ on the nose. Write the scores on the probability-integral scale, $U_i = F(S_i)$, so each U_i is marginally uniform whatever the model. With $k = \lceil (n+1)(1-\alpha) \rceil$ the conformal threshold is the order statistic $U_{(k)}$, and the calibration-conditional coverage against an independent test draw from the marginal is

$$c = U_{(k)}. \tag{1}$$

For independent (or merely exchangeable, de facto independent) calibration scores this is the classical result (Vovk, 2012; Marques F., 2024): $c \sim \mathrm{Beta}(k, n-k+1)$, with

$$\mathbb{E}[c] = \frac{k}{n+1}, \qquad \mathrm{Var}(c) = \frac{k(n-k+1)}{(n+1)^2(n+2)} \;\approx\; \frac{\alpha(1-\alpha)}{n}. \tag{2}$$

This is the fan: the per-dataset coverage scatters around $1-\alpha$ with a spread that shrinks like $1/n$. The question of this note is what the dependence among the calibration scores does to that spread. The companion notes treat the orthogonal, within-dataset axis (coverage uneven across x

for a fixed dataset, the information gap $I(R;X)$ and its sign Cotton; Cotton); here every statement is about the across-dataset fluctuation of the single number c.

The whole problem passes through the count process $N(u) = \sum_{i=1}^{n} \mathbf{1}\{U_i \le u\}$, because $U_{(k)} = \int_0^1 \mathbf{1}\{N(u) < k\}\,\mathrm{d}u$, so

$$\operatorname{Var}(c) = \int_0^1\int_0^1 \operatorname{Cov}\big(\mathbf{1}\{N(u) < k\},\, \mathbf{1}\{N(v) < k\}\big)\,\mathrm{d}u\,\mathrm{d}v, \tag{3}$$

a functional of the law of the count process alone. Two facts about N drive everything.

Lemma 1 (Count variance). *If the scores are negatively associated (Joag-Dev and Proschan, 1983), then for every u, $\operatorname{Var}(N(u)) \le n\,u(1-u)$, with equality iff the exceedance indicators are pairwise uncorrelated.*

Proof. The indicators $\mathbf{1}\{U_i \le u\}$ are coordinatewise-monotone functions of single distinct coordinates, hence negatively associated, hence pairwise non-positively correlated. So $\operatorname{Var}(N(u)) = \sum_i \operatorname{Var} + \sum_{i\ne j}\operatorname{Cov} \le \sum_i u(1-u) = n\,u(1-u)$. □

2 The fan coefficient

Let $p = 1-\alpha$ be the operating level and $\xi_i = \mathbf{1}\{U_i \le p\}$, so $N(p) = \sum_i \xi_i$. The Bahadur representation for the uniform quantile (Bahadur, 1966), which holds for negatively associated sequences under mild conditions, gives $U_{(k)} - p = p - N(p)/n + o_P(n^{-1/2})$, and hence

Theorem 1 (Fan coefficient). *As $n \to \infty$,*

$$\operatorname{Var}(c) = \frac{1}{n^2}\operatorname{Var}(N(p))\,(1+o(1)) = \frac{\alpha(1-\alpha)}{n}\,R\,(1+o(1)), \qquad R = 1 + \frac{n-1}{\alpha(1-\alpha)}\,\bar{c}, \tag{4}$$

where $\bar{c} = \frac{1}{n(n-1)}\sum_{i\ne j}\operatorname{Cov}(\xi_i,\xi_j)$ is the average pairwise covariance of the exceedance indicators at the operating quantile.

The fan's leading coefficient is, exactly, the variance of the exceedance count at $1-\alpha$. Under independence $\bar{c} = 0$ and $R = 1$, recovering (2). Negative association makes $\bar{c} < 0$ and $R < 1$: the fan narrows. At the contest floor $\bar{c} = -\alpha(1-\alpha)/(n-1)$ one gets $R = 0$. The reduction factor R is the single number to carry away.

3 Positive dependence widens the fan, exactly

When the scores are extendable, that is, drawn from an infinitely exchangeable sequence, de Finetti (de Finetti, 1937; Kerns and Székely, 2006) makes them conditionally independent given a latent directing measure Q. Conditional on Q the coverage c is an ordinary independent-sample order statistic, so the law of total variance gives an exact decomposition.

Theorem 2 (Positive dependence). *For an extendable exchangeable calibration sequence with directing measure Q,*

$$\operatorname{Var}(c) = \underbrace{\mathbb{E}_Q\big[\operatorname{Var}(c \mid Q)\big]}_{\textit{average within-dataset fan}} + \underbrace{\operatorname{Var}_Q\big(\mathbb{E}[c \mid Q]\big)}_{\textit{between-dataset term}}. \tag{5}$$

The between-dataset term is non-negative and vanishes under independence. Positive cross-sample dependence is exactly this term: a per-dataset latent that shifts the conditional quantile, on top of the ordinary fan.

The decomposition is exact and needs no asymptotics. It says the danger of positive dependence is not bias but variance: you draw a dataset, the latent Q places the whole calibration sample high or low, and the realized coverage rides with it. Numerically, a one-factor Gaussian model at pairwise correlation 0.2 has $\mathrm{Var}(c)$ about 3.3 times the independent fan, of which the between-dataset term is more than half.

4 Negative dependence narrows the fan

The negative regime is the one a genuine prior cannot reach. By Kerns and Székely (2006) a finite exchangeable sequence with $\rho < 0$ has a *signed* directing measure, so the clean conditioning of Theorem 2 is unavailable: there is no Q to condition on, and the between-dataset term would be a negative quasi-variance. That is precisely the regime where the fan contracts.

Theorem 3 (Endpoints)**.** *Independent scores give $c \sim \mathrm{Beta}(k, n-k+1)$, the widest fan at fixed marginals. At the maximally negatively associated floor, where the calibration scores are a uniformly random permutation of the grid $\{1/(n+1), \ldots, n/(n+1)\}$ (sampling without replacement, pairwise correlation $-1/(n-1)$), the order statistic is deterministic, $U_{(k)} = k/(n+1)$ almost surely, so $\mathrm{Var}(c) = 0$: the realized coverage equals the nominal level on every draw.*

Both endpoints agree with Theorem 1 at $R = 1$ and $R = 0$. Between them the fan contracts monotonically as the scores become more negatively dependent, and the contraction is exactly the $R < 1$ of the fan coefficient. The two endpoints are in fact joined by an exact path.

Proposition 1 (An exact path from independence to the floor)**.** *Let the calibration law be, with probability $1-t$, n independent uniforms, and with probability t, a contest (a uniformly random permutation of the grid $\{1/(n+1), \ldots, n/(n+1)\}$). Both components yield realized coverage with the same mean $k/(n+1)$, the contest with zero variance, so by the law of total variance the between-component term vanishes and*

$$\mathrm{Var}(c) = (1-t)\,\frac{k(n-k+1)}{(n+1)^2(n+2)}. \tag{6}$$

The pairwise correlation of the scores is $-t/(n-1)$, so as t runs from 0 to 1 the family runs from independence to the contest floor and the fan collapses linearly to zero.

Proof. Condition on the component. $\mathbb{E}[c \mid \text{iid}] = \mathbb{E}[c \mid \text{contest}] = k/(n+1)$, so $\mathrm{Var}(\mathbb{E}[c \mid \cdot]) = 0$; and $\mathbb{E}[\mathrm{Var}(c \mid \cdot)] = (1-t) \cdot \frac{k(n-k+1)}{(n+1)^2(n+2)} + t \cdot 0$. Add the two. The correlation is $(1-t) \cdot 0 + t \cdot (-1/(n-1))$ since both components share the mean. □

So we can prove the contraction exactly at both endpoints and along this whole interpolating family, and to leading order everywhere (Theorem 1). For an *arbitrary* negatively associated law the contraction also holds once it is summed over all levels.

Theorem 4 (Aggregate fan bound)**.** *For exchangeable, negatively associated scores with uniform marginals,*

$$\sum_{k=1}^{n} \mathrm{Var}(U_{(k)}) \;\le\; \sum_{k=1}^{n} \frac{k(n-k+1)}{(n+1)^2(n+2)}, \tag{7}$$

the independent value, with equality iff the scores are independent. Summed over all target levels, the fan is never widened by negative association.

Proof. The order statistics are a permutation of the sample, so $\sum_k U_{(k)}^2 = \sum_i U_i^2$ and, under uniform marginals, $\sum_k \mathbb{E}[U_{(k)}^2] = n\,\mathbb{E}[U_1^2] = n/3$ for every such law; hence $\sum_k \mathrm{Var}(U_{(k)}) = \frac{n}{3} - \sum_k (\mathbb{E}\,U_{(k)})^2$. For the top j statistics, $\sum_{k>n-j} U_{(k)} = \int_0^1 \min\big(j, n - N(u)\big)\,\mathrm{d}u$, and $m \mapsto \min(j, n-m)$ is concave. By Shao (2000) the exceedance count obeys $N(u) \le_{\mathrm{cx}} \mathrm{Bin}(n,u)$, so for a concave map the inequality reverses, $\mathbb{E}\min(j, n - N(u)) \ge \mathbb{E}\min(j, n - \mathrm{Bin}(n,u))$; integrating, $\mathbb{E}\big[\sum_{k>n-j} U_{(k)}\big]$ is at least its independent value for every j. As the totals agree ($\sum_k \mathbb{E}\,U_{(k)} = n/2$), the profile $(\mathbb{E}\,U_{(1)}, \dots, \mathbb{E}\,U_{(n)})$ majorizes the independent profile $(\frac{1}{n+1}, \dots, \frac{n}{n+1})$. Since $\sum_k x_k^2$ is Schur-convex, $\sum_k (\mathbb{E}\,U_{(k)})^2$ is at least its independent value, and the identity above finishes it. □

The per-level refinement, for the single conformal k, remains open, and the obstruction is exactly why: the step $\mathbf{1}\{N(u) < k\}$ behind a single $\mathrm{Var}(U_{(k)})$ is neither concave nor convex, so the convex order does not sign it, whereas the smooth aggregate functional $\min(j, n-N)$ is concave and it does.

Conjecture 1 (Per-level contraction)**.** *If the calibration scores are negatively associated then* $\mathrm{Var}(c)$ *is at most the independent value* (2) *for each k individually.*

In the canonical continuous family the per-level statement is exact, and more, it is monotone in the correlation.

Proposition 2 (Exact per-level law for equicorrelated normals)**.** *Let $(Z_1, \dots, Z_n)$ be equicorrelated standard normal with correlation $\rho \in [-\frac{1}{n-1}, 1]$, and let $v_k = \mathrm{Var}(\xi_{(k)})$ be the variance of the k-th order statistic of n i.i.d. standard normals. Then, for every k,*

$$\mathrm{Var}(Z_{(k)}) = v_k + \rho\,(1 - v_k). \tag{8}$$

As $v_k < 1$ for $n \ge 2$, this is strictly increasing in ρ: negative correlation strictly shrinks the variance of every order statistic, positive correlation inflates it.

Proof. Write $Z_i = \bar{Z} + (Z_i - \bar{Z})$. For equicorrelated normals $\bar{Z} \sim N(0,s)$, $s = (1 + (n-1)\rho)/n$, is independent of the centred vector, and $\mathrm{Cov}(Z_i - \bar{Z}, Z_j - \bar{Z}) = (1-\rho)(\delta_{ij} - 1/n)$, exactly $1-\rho$ times the covariance of $(\xi_i - \bar{\xi})$ for i.i.d. standard normals. Hence $Z_{(k)} \overset{d}{=} \sqrt{s}\,\eta + \sqrt{1-\rho}\,(\xi - \bar{\xi})_{(k)}$ with $\eta \sim N(0,1)$ independent of the second term, so $\mathrm{Var}(Z_{(k)}) = s + (1-\rho)\,\mathrm{Var}\big((\xi - \bar{\xi})_{(k)}\big)$. Both terms are affine in ρ; matching v_k at $\rho = 0$ forces $\mathrm{Var}((\xi - \bar{\xi})_{(k)}) = v_k - 1/n$, and substituting gives $v_k + \rho(1 - v_k)$. □

Remark 1. *This is the per-level contraction, exact and monotone, on the Gaussian scale, unifying the two sides: the single line $v_k + \rho(1-v_k)$ runs from inflation at $\rho > 0$ through the independent value at $\rho = 0$ to the floor at $\rho = -1/(n-1)$. Under the Gaussian copula $U_i = \Phi(Z_i)$, which restores uniform marginals,* $\mathrm{Var}(U_{(k)})$ *is numerically monotone in ρ and below the independent value throughout $\rho < 0$ (`check_fan.py`). The transfer to the uniform scale is a dispersive-order statement, and it is half rigorous. Decompose along the common mean, $Z_{(k)} = \sqrt{1-\rho}\,D + \bar{Z}$ with $D = (\xi - \bar{\xi})_{(k)}$ and $\bar{Z} \sim N(0,s)$ independent. D is log-concave, being the order statistic of a vector with log-concave (Gaussian) joint density (Prékopa, 1973); and convolution with a log-concave density increases the dispersive order (Lewis and Thompson, 1981), so adding the Gaussian shift $\bar{Z}$ disperses, monotonically in s. The residual is that this outweighs the simultaneous $\sqrt{1-\rho}$ shrinkage of D as ρ grows, a one-parameter density-quantile monotonicity under the variance budget $\frac{n-1}{n}(1-\rho) + s = 1$, which no off-the-shelf convolution theorem delivers but which we now settle.*

Tweedie's formula sharpens that residual to a single posterior-variance bound. Reading $Y := Z_{(k)} = aD + b\eta$ ($a = \sqrt{1-\rho}$, $b = \sqrt{s}$) as Gaussian signal extraction of the log-concave signal D, *the slope of the posterior mean equals the scaled posterior variance,* $\frac{d}{dq}\mathbb{E}[D \mid Y = q] = a\,\mathrm{Var}(D \mid Y = q)/b^2$ *(Efron, 2011; Hansen and Tong, 2026). The dispersive monotonicity is equivalent to this slope being at most* $(n-1)a/n$*, so, using* $\frac{n-1}{n}a^2 + b^2 = 1$*, the entire transfer reduces to*

$$\mathrm{Var}(D \mid Y = q) \;\leq\; \frac{n-1}{n}\,b^2 \qquad \textit{for all } q, \tag{9}$$

a posterior-variance bound for log-concave Gaussian denoising, which the next lemma settles. The score-driven reading of the same identity (Hansen and Tong, 2026) is the econometric form of modelling the conditional spread, which is exactly what closes the information gap a conformal wrapper leaves (Cotton).

Lemma 2 (Strong log-concavity of D)**.** *$D = (\xi - \bar{\xi})_{(k)}$ is $\frac{n}{n-1}$-strongly log-concave,* $(\log g_D)'' \leq -n/(n-1)$.

Proof. The centred vector $w = \xi - \bar{\xi}$ has covariance $I - \frac{1}{n}J$, an orthogonal projection, so on the hyperplane $\{\sum_i w_i = 0\}$ it is a *standard* Gaussian: its $-\log$ density is $\frac{1}{2}\|w\|^2$, with Hessian the identity. The law of the ordered vector is this density restricted to the convex cone $\{w_1 \leq \cdots \leq w_n\}$, which adds only a convex term to $-\log$ and so stays strongly log-concave with at least the same constant. Strong log-concavity is preserved under marginalization (Prékopa, 1973; Brascamp and Lieb, 1976; Saumard and Wellner, 2014), with the marginal of a coordinate strongly log-concave to the reciprocal of that coordinate's reference variance; here the coordinate $w_{(k)}$ has reference variance $\frac{n-1}{n}$ (the diagonal of $I - \frac{1}{n}J$), giving constant $\frac{n}{n-1}$. □

Proposition 3 (Gaussian-scale dispersive monotonicity)**.** *For equicorrelated standard normals, $Z_{(k)}$ is increasing in the dispersive order as ρ increases.*

Proof. The posterior of D given $Y = aD + b\eta$ has $-\log$ density with Hessian $(-\log g_D)'' + a^2/b^2 \geq \frac{n}{n-1} + \frac{a^2}{b^2}$ by Lemma 2, and the budget $\frac{n-1}{n}a^2 + b^2 = 1$ makes $\frac{n}{n-1} + \frac{a^2}{b^2} = \frac{n}{(n-1)b^2}$. The Brascamp–Lieb variance inequality (Brascamp and Lieb, 1976) then gives $\mathrm{Var}(D \mid Y = q) \leq (n-1)b^2/n$, the bound above, so $\frac{d}{dq}\mathbb{E}[D \mid Y = q] \leq (n-1)a/n$ and the quantile density of $Z_{(k)}$ decreases in ρ, which is the dispersive monotonicity. □

This settles the Gaussian-scale per-level statement in the strong (dispersive, not merely variance) sense. The one step left for the uniform-marginal conjecture under the Gaussian copula is the passage through Φ: being S-shaped, Φ is neither convex nor concave, so it need not transmit the dispersive order, and $\mathrm{Var}(U_{(k)}) = \mathrm{Var}(\Phi(Z_{(k)}))$ monotone in ρ is confirmed numerically (`check_fan.py`) but not by a transfer theorem.

The same convex-order contraction $N(u) \leq_{\mathrm{cx}} \mathrm{Bin}(n, u)$ that drives the aggregate bound is available per level, but turning it into the single-k statement needs the tail $\mathbb{P}(N(u) \geq k)$ to cross the binomial tail once in u, which the total positivity of the binomial tail in (k, u) (Karlin, 1968) makes plausible, followed by a single-crossing-implies-less-spread step. We claim only the variance statement, and deliberately not the stronger dispersive ordering: an order statistic of a dependent sample need not be dispersively smaller than the independent one even at fixed marginals (Panja et al., 2022), so the result is specific to the variance and is not a corollary of a dispersive theorem. The natural sufficient condition on the dependence is negative supermodular dependence (Christofides and Vaggelatou, 2004), which all the structures below satisfy. We verify the per-level conjecture across them (one run each, reproduced by `check_fan.py`; $n = 20$, $\alpha = 0.1$, independent fan variance 3.92×10^{-3}).

calibration dependence	$\mathbb{E}[c]$	$\mathrm{Var}(c)$	ratio to fan	sign
independent ($\rho = 0$)	0.904	3.92×10^{-3}	1.00	—
exchangeable, $\rho = -1/(n-1)$	0.923	1.16×10^{-3}	0.29	neg. assoc.
MA(1), lag-1 corr $-\frac{1}{2}$	0.912	2.71×10^{-3}	0.69	neg. assoc.
random negatively-paired blocks	0.909	3.14×10^{-3}	0.80	neg. assoc.
contest / without replacement	0.905	0	0.00	floor
one-factor, $\rho = +0.2$	0.863	1.33×10^{-2}	3.41	positive

Every negatively associated structure, exchangeable or not, comes in under the independent fan; the positive control widens it. The mean drifts up slightly under negative dependence and down under positive dependence, but stays near $1-\alpha$; the action is in the variance. A wider search agrees: over exchangeable, moving-average, block, and randomly-generated non-exchangeable negatively-correlated Gaussian copulas at $n \le 8$ and every k, the ratio to the independent fan never exceeds one (it ranges from about 0.30 at the floor to 0.80 for weak dependence), with no counterexample. In all of these the order-statistic CDFs $\mathbb{P}(N(u) \ge k)$ cross the independent ones exactly once, the single crossing the remark below uses.

Remark 2 (A closed form, and what an exact proof must dominate)**.** *The variance is an explicit functional of $g(u) = \mathbb{P}(N(u) \ge k)$,*

$$\mathrm{Var}(c) = 2\int_0^1 (1-u)\, g(u)\,\mathrm{d}u - \Big(\int_0^1 g(u)\,\mathrm{d}u\Big)^2, \tag{10}$$

so writing $\delta = g - g^{\mathrm{iid}}$, $A = \int_0^1 g^{\mathrm{iid}} = 1 - \mathbb{E}[c^{\mathrm{iid}}]$, and $\Delta = \int_0^1 \delta$,

$$\mathrm{Var}(c) - \mathrm{Var}(c^{\mathrm{iid}}) = 2\int_0^1 \big[(1-u) - A\big]\,\delta(u)\,\mathrm{d}u - \Delta^2. \tag{11}$$

The $-\Delta^2$ term is non-positive. The weight $(1-u) - A$ changes sign at $u = \mathbb{E}[c^{\mathrm{iid}}] = k/(n+1)$, and under negative association δ changes sign once, at the level $u \approx k/n$ where the operating quantile crosses the count mean and the contraction of Lemma 1 flips the tail comparison. The two sign changes agree up to the conformal offset between k/n and $k/(n+1)$, a band of width $O(1/n)$ on which the integrand is positive; everywhere else it is negative. So the identity proves the contraction up to that single $O(1/n)$ boundary term, a second route to the leading order of Theorem 1, and isolates exactly what a finite-sample proof must control: the boundary remainder against the slack Δ^2.

Remark 3 (Why the closed form does not close it, and the lemma that would)**.** *The closed form is, by itself, circular. Substituting the moment identities $\int u\,\delta = \frac{1}{2}(\mathbb{E}[c_{\mathrm{iid}}^2] - \mathbb{E}[c^2])$ and $\int \delta = \mathbb{E}[c_{\mathrm{iid}}] - \mathbb{E}[c]$ into any elementary rearrangement of the difference, single-crossing splits, Abel summation, the exact first-order perturbation, collapses it back to $\mathrm{Var}(c) - \mathrm{Var}(c_{\mathrm{iid}})$ with the auxiliary crossing point cancelling: these structural facts carry no magnitude. Nor is the per-level convex order of Lemma-1 enough on its own. Because $\mathrm{Var}(c)$ depends only on the single curve $g(u) = \mathbb{P}(N(u) \ge k)$, one might hope $N(u) \le_{\mathrm{cx}} \mathrm{Bin}(n,u)$ at each u suffices; it does not, since a curve dominated pointwise can have larger variance (place an atom at the boundary), and such curves are excluded only because they are not realizable by a genuine sample. The content is therefore an extremal statement about the realizable curves, not their marginals:*

> *Open lemma. Among all $g(u) = \mathbb{P}(N(u) \ge k)$ realizable by a negatively associated uniform sample, the independent (binomial) curve maximizes $\mathrm{Var}(c)$.*

This is what a total-positivity or coupling argument would have to deliver. Proposition 1 and the contest endpoint are the cases settled by hand.

Remark 4 (The signed measure as a negative between-term)**.** *Theorem 2 reads the fan as within-dataset variance plus a non-negative between-dataset variance. In the signed-de-Finetti picture of the companion note, the negative corner is exactly where that between-dataset object turns negative, a Feynman–Wigner quasi-variance that subtracts from the within-dataset fan rather than adding to it, reaching the full cancellation $R = 0$ at the contest floor. This is heuristic, since a signed measure has no honest conditional variance, but it is the right cartoon: the same sign that decides conditional adaptivity in Cotton decides whether the fan is inflated or cancelled here.*

5 A constructive corollary

If negative dependence narrows the fan, one can engineer it. A balanced calibration set whose scores behave like the grid of Theorem 3 concentrates the realized coverage far inside the independent fan, so the same reliability is bought with fewer calibration points, or the same number of points gives less run-to-run wobble. The classical variance-reduction designs all apply in principle: stratification never increases variance (Cochran, 1977), antithetic pairing helps for the monotone empirical CDF, and determinantal or repulsive designs give negative association with faster-than-independent concentration (Hough et al., 2006; Bardenet and Hardy, 2020).

Remark 5 (The exchangeability constraint)**.** *The marginal guarantee needs each calibration score exchangeable with the test score; it does not need the calibration scores independent of one another. So negative association is admissible, but only if it is induced by a map symmetric across all $n + 1$ points, a pooled de-meaning or ranking, which preserves the exchangeability of the joint. Selecting or curating the calibration set by its scores, the obvious way to balance it, breaks exchangeability with the test point and voids the guarantee. The viable construction is transductive and symmetric; that is also why the contest grid, a symmetric function of the pooled sample, both keeps the marginal guarantee and zeroes the fan.*

6 Related work

The Beta$(k, n - k + 1)$ law of calibration-conditional coverage and its variance are Vovk (2012) and Marques F. (2024); the entire prior literature treats the calibration scores as independent or exchangeable and reads off a single number, never varying the dependence to move it. Negative association and its consequences are Joag-Dev and Proschan (1983); that negatively associated sums are dominated in convex order by independent ones is Shao (2000), and sampling without replacement concentrates at least as tightly as with replacement (Serfling, 1974). The total positivity behind the single-crossing step is Karlin (1968); that an order statistic of a dependent sample need not be dispersively smaller than the independent one, which is why we claim only the variance, is Panja et al. (2022); the connection between supermodular ordering and negative dependence is Christofides and Vaggelatou (2004); the Bahadur representation is Bahadur (1966). Determinantal and repulsive designs are Hough et al. (2006); Bardenet and Hardy (2020) and stratification is Cochran (1977). The finite signed de Finetti representation is Kerns and Székely (2006), and the within-dataset, across-x reading of its sign is the companion note (Cotton); the information gap is (Cotton). The contribution here is to make the dependence the control variable for the *variance* of realized coverage: the exact fan coefficient R, the exact de Finetti decomposition for the positive side, the exact zero at the contest floor, and the reduction of the general negative case to a dispersive ordering.

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